

EFFECTIVE POROSITY AND FLOW RATE WITH INFILTRATION AT LOW TENSIONS INTO A WELL-STRUCTURED SUBSOIL

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ABSTRACT. A combination of tension infiltrometers and dye tracers was used to investigate the extent and nature of water movement at low tensions through the well-structured subsoil of Ships clay (very-fine, mixed, thermic, Chromic Hapluderts). Dyed-water was placed in the reservoirs of tension infiltrometers and allowed to infiltrate into the soil under seven tension sequences to separate different size pores effective in transmitting water. Flow patterns were then revealed by exposing dye-stained soil after each infiltration sequence. The results showed that, when water was supplied at tensions <24 cm, the effective porosity (fraction of pores that were stained) in this subsoil was primarily macro- and mesoporosity, which constituted about 5% of the total soil porosity. Under flow at 0-cm tension, macropores ≥ 0.5 mm and mesopores from 0.06 to 0.5 mm (radius for cylindrical pores or width for planar pores) contributed about 89% and 10% of the total water flux, respectively. Micropores <0.06 mm contributed the remaining 1% of the total water flux, but constituted about 95% of the total soil porosity. Dye stain patterns showed that water flow at tensions <24 cm in this structured subsoil was primarily controlled by slickenside fissures, root channels, and vertical fissures. The nonuniform flow pattern caused water to penetrate to depths as much as 11 times deeper than expected from a Green-Ampt model that considered the total soil porosity to be active in transmitting water. By replacing the air-filled porosity in the Green-Ampt model with the active macro- plus mesoporosity, calculated wetting front depths were close to the observed maximum dye depths. **Keywords.** Macropore flow, Fractional wetting, Structured soil, Dye-tracing, Tension infiltrometer.

Water movement in naturally structured soils is different from water movement in artificially homogenized soil. The presence of interpedal and biological pores in natural soils leads to spatial concentration of water fluxes (Beven and Germann, 1982; Booltink and Bouma, 1991) such that a considerable amount of the soil matrix is by-passed as water moves through it (Ghodrati and Jury, 1990; Lin and McInnes, 1995). The fraction of total soil porosity that contributes to water flow influences the depth that water penetrates after infiltration. The smaller the fraction of porosity transmitting water, the faster water moves as it travels through the soil.

Tension infiltrometers provide a useful way of separating water movement through large interpedal and biological pores from water movement through the soil matrix (Perroux and White 1988; Ankeny et al., 1988; Elrick et al., 1987). By permitting water to be supplied for infiltration at a tension that excludes pores larger than a given diameter or width from participating in flow, the

infiltrometers allow for an easy quantification of the contribution of different pore sizes to water flow (Watson and Luxmoore, 1986; Jarvis et al., 1987; Messing and Jarvis, 1993).

Our objective was to define how water moved through the subsoil of a well-structured vertisol. We combined the use of a tension infiltrometer and a dye tracer to mark the water flow paths so that the extent and nature of the water movement in the subsoil could be described. We comment on porosity effective in transmitting water and on infiltration rate.

MATERIALS AND METHODS

The subsoil of a representative vertisol, Ships clay (very-fine, mixed, thermic Chromic Hapluderts), located on the Brazos River flood plain near College Station, Texas, was selected for this study. We chose the subsoil because we had preliminary evidence that some of the water passing through the surface soil moved quickly through the subsoil. This rapid movement through the subsoil could lead to contaminants quickly reaching groundwater. The research site was fallow at the time the experiments were conducted. The soil above the subsoil consisted of a tilled Ap horizon (0 to 10 cm) and a dense plowpan (10 to 27 cm). The subsoil had a Bss horizon from 27 to 122 cm with a clay texture (70% clay and <1% sand). The Bss horizon had moderate coarse angular blocky parting to moderate fine angular blocky pedality, few to common very fine intrapedal cylindrical pores (mostly root channels about 1 mm diameter), and common very fine planar fissures between slickenside planes (generally <1 mm in width and tilted 30 to 60° from horizontal). Below 122 cm, there were stratified layers of silty clay

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loam (BC horizon) with weak to moderate fine and very fine angular blocky pedality and massive silt loam (2C horizons).

Infiltration and dye-tracing measurements were made in the Bss horizon about 50 cm below the soil surface. At this depth, the subsoil had an unconfined bulk density at saturation of 1.24 Mg m^{-3} , a coefficient of linear extensibility of 0.16 m m^{-1} , a pH of 7.8, an organic carbon content of 7.7 g kg^{-1} , and a calcium carbonate content of 133 g kg^{-1} . Infiltration surfaces in the subsoil were exposed on horizontal planes at least 0.5 m inward from the walls of freshly dug pits. Each infiltration surface exposed for measurement was prepared by gently plucking small pieces of soil from the surface until it was level. This was done to preserve natural soil pedality and porosity. At the time of this study, the subsoil was moist from heavy rainfall events the preceding weeks. Water contents of the soil beneath the surfaces, determined gravimetrically from samples taken before the infiltration measurements, ranged from 0.33 to 0.37 kg kg^{-1} .

A soil water desorption curve was determined using both natural clods and core samples placed on porous ceramic plates in pressure chambers (Klute, 1986; Soil Survey Laboratory Staff, 1992). Four clods and three cores, each of about 350 cm^3 in size, were collected from the field site and presaturated before determining water retention by desorption. The water retention data were used to estimate soil pore-size distribution. For comparison with field tension infiltration data, soil macropores were classified as the pores active at tensions $\leq 3 \text{ cm}$, mesopores as those active from 3 to 24 cm tension, and micropores as those active at tensions $> 24 \text{ cm}$. This pore classification scheme was different than that defined by Luxmoore (1981) in that the tension separating mesopores and micropores was 24 cm instead of 300 cm.

Infiltrimeters following the design of Perroux and White (1988) with circular infiltration surfaces 24.5 cm diameter were used in this study. Water-soluble anionic dyes, either Standard Blue or FWT Red (Formulabs, Inc., Piqua, Ohio), were used for tracing the extent and pattern of macropore flow. Compared to other dyes tested, both dyes had high visibility, high solubility, and low adsorption. No readily discernible difference in movement between Standard Blue and FWT Red dyes were observed in preliminary tests. There was no reason for using the two different dyes other than a continued observation of their flow and staining characteristics. Aqueous solutions of 0.5% Standard Blue or 1% FWT Red were placed in the water reservoirs of tension infiltrimeters and supplied to the soil at a sequence of controlled tensions. We operated the infiltrimeters with the supply water potential at the infiltration surface in the following seven sequences: 24 cm; 24, 12 cm; 24, 12, 6 cm; 24, 12, 6, 3 cm; 24, 12, 6, 3, 2 cm; 24, 12, 6, 3, 2, 1 cm; and 24, 12, 6, 3, 2, 1, 0 cm. The water was supplied at tensions 24, 12, 6, 3, 2, 1, and 0 cm for 30, 20, 20, 10, 10, 10, and 10 min, respectively. From the capillary rise equation, supply water tensions of 24, 12, 6, 3, 2, and 1 cm should have excluded cylindrical pores of radius, or planar pores of width, greater than 0.0625, 0.125, 0.25, 0.5, 0.75, and 1.5 mm, respectively, from participating in flow. A supply tension of 0 cm should have allowed all pores to transmit water. Measurements in the seven sequences were duplicated, once with Standard

Blue and once with FWT Red. Contributions of different size pores to the total flow were made by differences between individual sequences. To further investigate the relative contributions of different types of flow paths to the overall flow, a prolonged infiltration was conducted for a total of about 19 h.

Steady infiltration rate, $i_h \text{ (m s}^{-1}\text{)}$, was expressed as water flux per unit area of infiltrimeter surface, $i_h = Q_h/A$, where $Q_h \text{ (m}^3 \text{ s}^{-1}\text{)}$ is the apparent steady-state flux at supply tension h , and $A \text{ (m}^2\text{)}$ is the area of the infiltrimeter base. Contributions to total water flux by different pore size ranges were calculated as $\Delta Q/Q_0$, where ΔQ is the difference of apparent steady-state flux between two tensions (two pore size limits), and Q_0 is the apparent steady-state flux at 0-cm tension.

Immediately after an infiltration sequence was completed, the soil was excavated to examine the flow pattern and flow-related soil morphological features in both vertical and horizontal planar sections as shown by the dye stain. The downward flow pattern was revealed by cutting a vertical soil section along the central line of the intake area. Using the other half of the circular intake area, horizontal flow patterns were recorded at 0.5, 1, 5, and 10 cm below the infiltration surface, and at subsequent 10-cm increments until no dye stain was found. The observed maximum dye stain depth was recorded in each measurement. Accurate traces of the stained dye patterns and their relationship to soil morphological features on the exposed vertical/horizontal sections were made on clear plastic sheets. The information on plastic sheets was then digitized into a computer database through PC ARC/INFO (Environmental System Research Institute, Inc., Redlands, Calif.) for later reproduction.

To compare theoretical and measured infiltration, maximum wetting front depth in each infiltration sequence was calculated based on the Green-Ampt infiltration model. The depth of the wetting front $L_F \text{ (m)}$ was calculated (Skaggs and Khaleel, 1982; Marshall and Holmes, 1988) as:

$$L_F = V_t / (A \times \phi_{\text{air}}) \quad (1)$$

where

$$\begin{aligned} V_t \text{ (m}^3\text{)} &= \text{total volume of water infiltrated} \\ A \text{ (m}^2\text{)} &= \text{area of the infiltrimeter base} \\ \phi_{\text{air}} \text{ (m}^3 \text{ m}^{-3}\text{)} &= \text{air-filled porosity} \end{aligned}$$

Air-filled porosity was calculated as the difference between the total porosity and the volumetric water content. Because bases of the infiltrimeters we used were not large in size, flow was not one-dimensional. However, for short time periods and when gravity dominates flow, the assumption of one-dimensional flow is reasonable (Philip, 1968).

RESULTS AND DISCUSSION

As larger pores were allowed to transmit water, flow rate increased rapidly (fig. 1). On average, the ratios of macropore flow rate ($i_0 - i_3$), mesopore flow rate ($i_3 - i_{24}$), and micropore flow rate (i_{24}) were 92:11:1. The water retention curve, based on pressure plate analysis, showed that the soil studied had $0.015 \text{ m}^3 \text{ m}^{-3}$ macroporosity, $0.010 \text{ m}^3 \text{ m}^{-3}$ mesoporosity, and $0.515 \text{ m}^3 \text{ m}^{-3}$

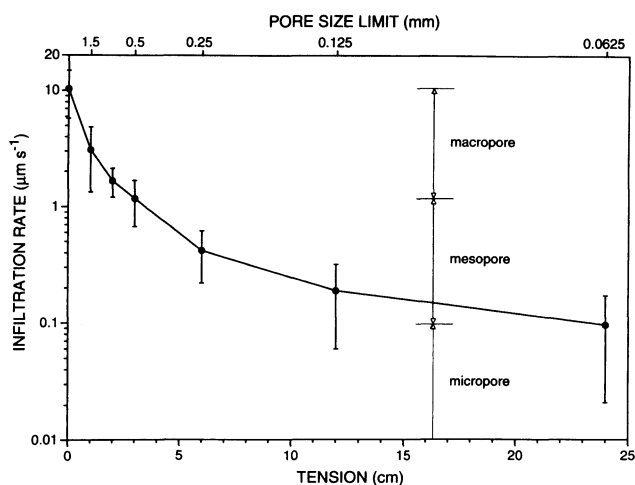


Figure 1—Apparent steady infiltration rate as a function of supply tension in the subsoil of Ships clay. Points are the average values and error bars indicate $\pm$ S.D. ($n = 14$ at 24 cm and $n = 2$ at 0 cm). Pore size describes radius for cylindrical pores and width for planar pores.

microporosity (fig. 2). Micropores dominated the total porosity of the soil, but transmitted little water in comparison to the macro- and mesopores. Comparison of soil pore volume fractions and the fractions of Q_0 conducted by macro-, meso-, and micropores showed that macropores contributed $88.7\% \pm 4.8\%$ (mean $\pm$ S.D.) of the

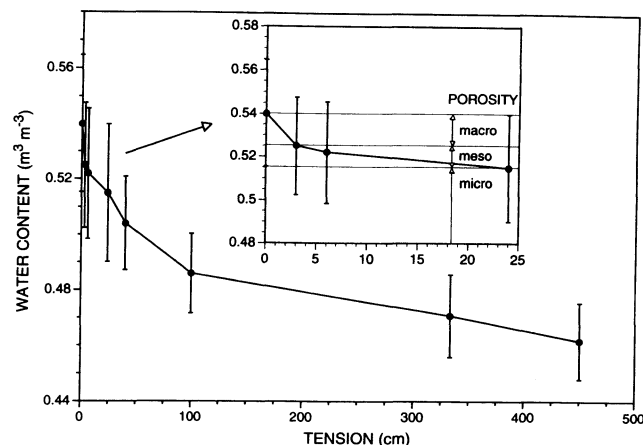


Figure 2—Water desorption curve of the soil studied. Points are the average values and error bars indicate $\pm$ S.D. ($n = 7$). Tension separating mesopores and micropores was 24 cm to facilitate comparisons with infiltration measurements.

total water flux, although macropores constituted only $2.8\% \pm 2.1\%$ of the total soil porosity. Mesopores constituted $1.8\% \pm 1.1\%$ of the total soil porosity, but contributed $10.4\% \pm 4.1\%$ of the total water flux. Micropores constituting $95.4\% \pm 4.6\%$ of the total soil porosity (about 12% were air-filled at the initiation of infiltration), contributed only $0.9\% \pm 0.7\%$ of the total water flux.

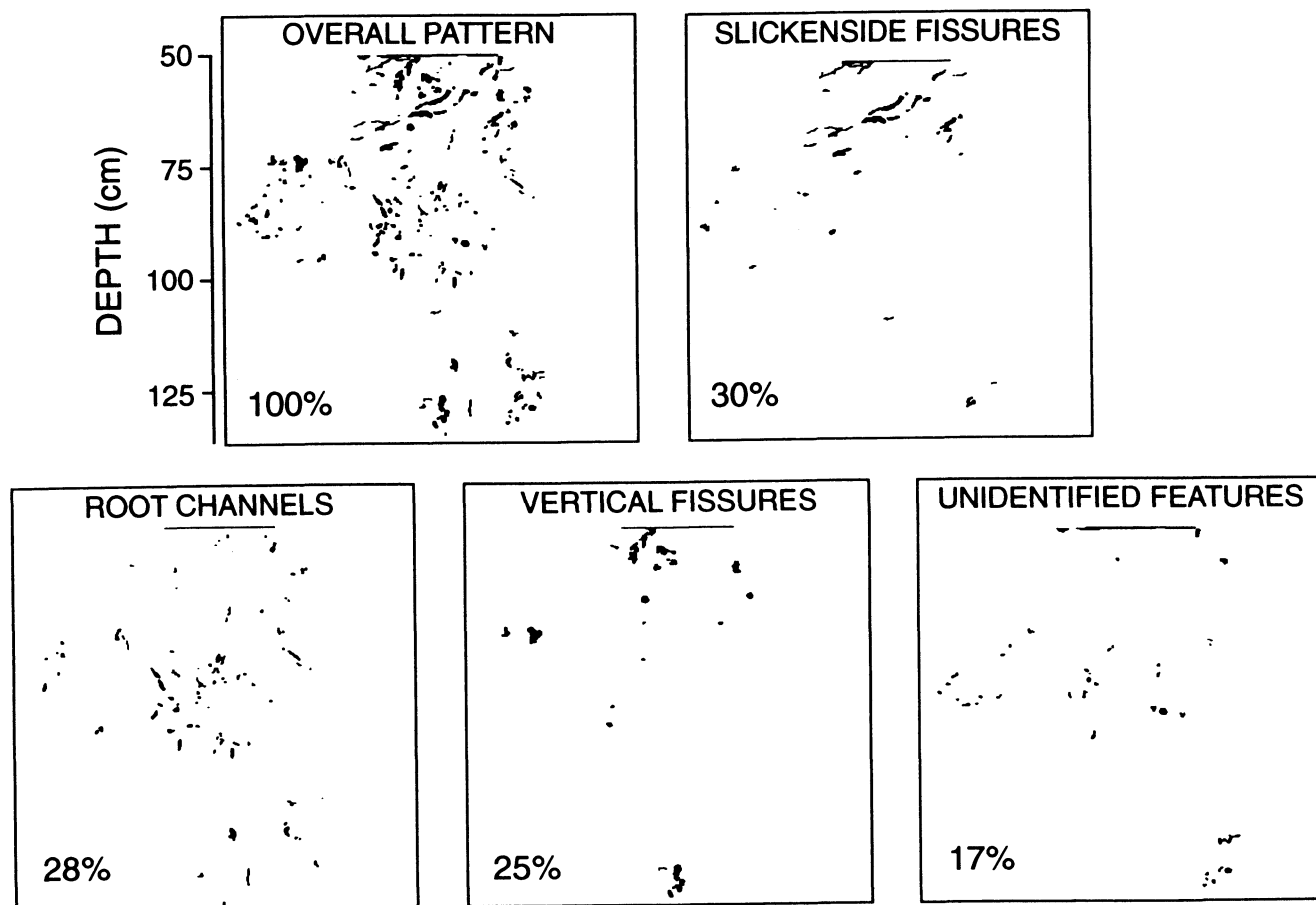


Figure 3—Contribution of slickenside fissures, root channels, vertical fissures, and unidentified features to the flow pattern exposed along the central line of the intake area after the prolonged infiltration. The horizontal lines in each subfigure indicate the locations of the infiltration intake areas (24.5 cm diameter). Vertical and horizontal scales are the same.

Field observations of vertical and horizontal dye stains exposed in the soil after the prolonged infiltration showed that effective flow paths at low tensions in this structured clayey soil were mainly macro- and mesopores associated with slickenside fissures, root channels, and vertical fissures (fig. 3). Dye stains exposed in the soil after each of the seven sequences showed similar results (fig. 4). When water was supplied at 12 or 6 cm tension, little dye stain was observed beneath the infiltration surface. Dye-stained area increased sharply when supply tensions were < 3 cm, where macropores were active. The rates of production of dye-stained areas relative to that at 0-cm supply tension were similar to the relative infiltration rates. For example, at 3-cm tension the relative rate of production of dye-stained area was 0.14 and the relative infiltration rate was 0.11. As expected, this suggested that the dye-stained area was proportional to the volume of water infiltrated.

The Green-Ampt infiltration model assumes a one-dimensional piston flow with a sharp wetting front (Skaggs and Khaleel, 1982; Jury et al., 1991). We did not observe a sharp wetting front. The flow patterns we measured at low tensions were primarily controlled by the distribution (including continuity and connectivity) of macro- and mesopores and thus were not one dimensional in nature. Because of the intersecting nature of obliquely oriented slickensides, scattered distribution of vertically or obliquely oriented root channels, and vertically oriented desiccation fissures, the wetting fronts in the soil studied were not well defined (figs. 3 and 4).

Because infiltration at low tensions in the soil studied advanced mostly along slickenside fissures, root channels, vertical fissures, and other macro- or mesopores, water flow generally by-passed most of the soil matrix. This permitted faster downward water movement and deeper penetration than was predicted from the Green-Ampt model; the observed maximum dye stain depths in the seven sequences were 1.1 to 11 times deeper than the predicted depths. Had we accounted for horizontal and not just vertical movement of water in the Green-Ampt model, differences between measured and predicted depths would have been larger.

Philip (1983) pointed out that in many natural and man-made situations, water infiltrates through only a fraction of the total soil surface (fractional wetting). Philip proposed the wetted fraction β (the fraction of the total surface area which is wetted) as a descriptor for the pattern of fractional wetting. With infiltration ending at the supply tension of 0 cm, the value of β with respect to A was < 0.1 , and ranged from 0.03 to 0.09. By replacing A in equation 1 with βA , calculated wetting front depths for infiltrations ending at the supply water tensions of 3 and 0 cm were closer to the observed maximum dye depths. However, because adsorption-diffusion processes occurred at macropore-matrix interfaces during the infiltration, the value of β contained some adsorptive matrix areas. When ϕ_{air} in equation 1 was replaced with the laboratory estimated fraction of the soil's macro- plus mesoporosity

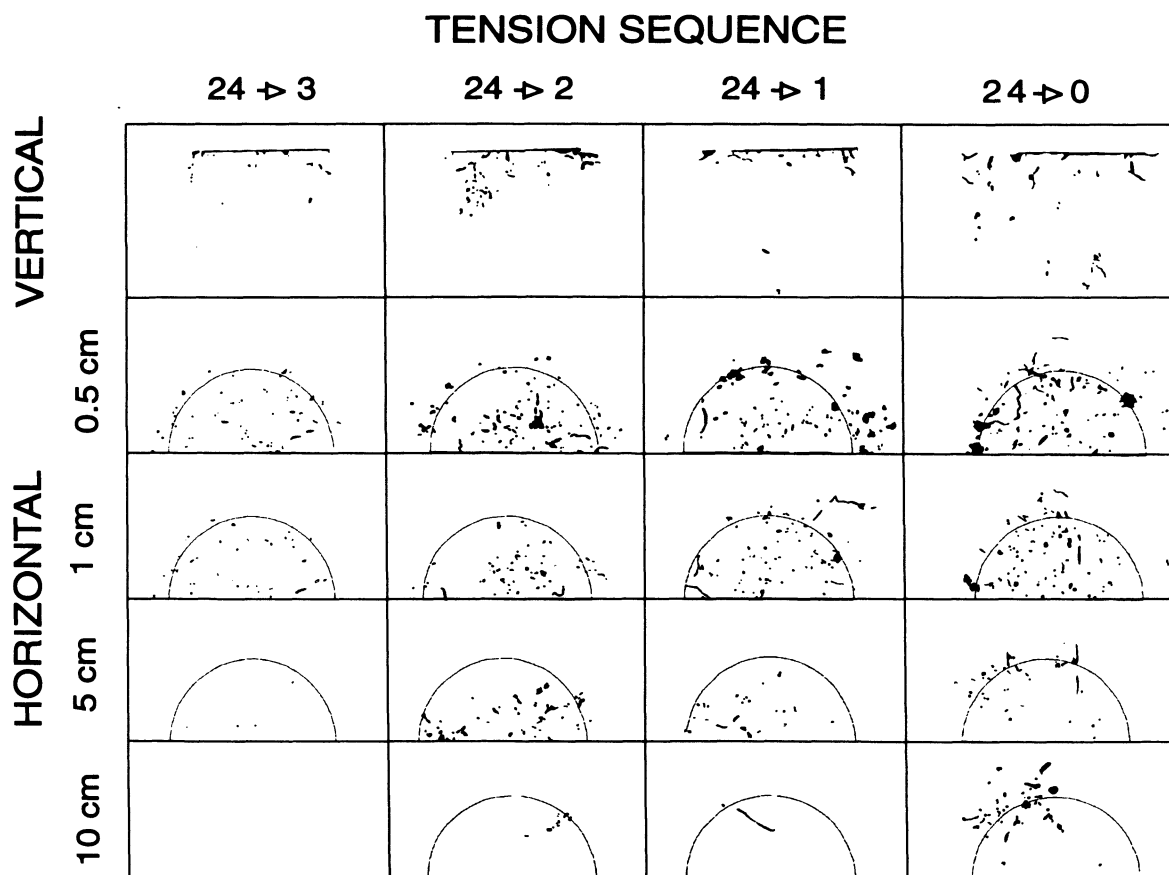


Figure 4—Representative dye stain patterns on vertical and horizontal sections under the infiltrometer for four of the seven tension sequences. The top lines in vertical sections and the semicircles (24.5 cm diameter) in horizontal sections indicate the corresponding locations of the infiltration intake areas. Vertical and horizontal scales are the same.

$\phi_{\text{macro+meso}}$ the resulting wetting front depths became very close to the observed maximum dye depths (fig. 5).

In summary, almost all of the total water infiltrated into the subsoil of Ships clay was transmitted through macro- and mesopores. Dye stain patterns indicated that water advanced mainly along slickenside fissures, root channels, vertical fissures, and other unidentified macro- or mesopores. This resulted in nonuniform flow patterns and water penetration much deeper than predicted with the standard Green-Ampt model. Deep penetration could be accounted for by considering that only macro- and mesopores transmitted water.

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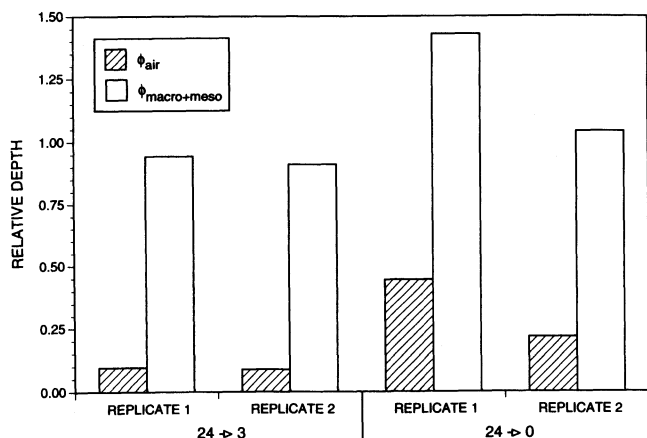


Figure 5—Depth of the Green-Ampt wetting front relative to the observed maximum dye depth for two measures of porosity. The closer the ratio is to 1, the better is the prediction.

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